ADJUNCTION TYPE BIJECTIONS FROM A LEFT DUAL IN A MONOIDAL CATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 9569-9600, following the statement at 9557-9567.

Editorial note. The omitted second equality (line 9599) is the epsilon-eta mirror of the fully executed first equality; retained verbatim as a mirror-level omission and covered by the card's mirror obligation.

Editorial note. The dual pair (Y, X) with eta and epsilon is fixed by definition-dual immediately preceding the statement.

Allowed standard prerequisites (editorial).

- Definition of a left dual. definition-dual immediately preceding the statement (source lines 9526-9550): eta: 1 -> X tensor Y and epsilon: Y tensor X -> 1 satisfying the two triangle diagrams. Supplies the unit and counit maps used to build the displayed composite maps at proof lines 9570-9583 and the identity (id tensor epsilon) after (eta tensor id) equals id used at proof lines 9596-9598.
- Bifunctoriality of the tensor product. the monoidal structure (C, tensor, phi) fixed in definition-dual. Makes the displayed composites of proof lines 9570-9583 and 9587-9592 well defined as maps of Hom sets, functorially in Z and Z'.

Lemma 0.1. *Let $\mathcal{C}$ be a monoidal category. If Y is a left dual to X, then*

$$\mathrm{Mor}(Z' \otimes X, Z) = \mathrm{Mor}(Z', Z \otimes Y) \quad \textit{and} \quad \mathrm{Mor}(Y \otimes Z', Z) = \mathrm{Mor}(Z', X \otimes Z)$$

functorially in Z and Z'.

Proof. Consider the maps

$$\mathrm{Mor}(Z' \otimes X, Z) \to \mathrm{Mor}(Z' \otimes X \otimes Y, Z \otimes Y) \to \mathrm{Mor}(Z', Z \otimes Y)$$

where we use η in the second arrow and the sequence of maps

$$\mathrm{Mor}(Z', Z \otimes Y) \to \mathrm{Mor}(Z' \otimes X, Z \otimes Y \otimes X) \to \mathrm{Mor}(Z' \otimes X, Z)$$

where we use ϵ in the second arrow. To show these arrows are mutually inverse, consider a map $a : Z' \to Z \otimes Y$. We have to show that

$$Z' \xrightarrow{\mathrm{id}_{Z'} \otimes \eta} Z' \otimes X \otimes Y \xrightarrow{a \otimes \mathrm{id}_{X \otimes Y}} Z \otimes Y \otimes X \otimes Y \xrightarrow{\mathrm{id}_Z \otimes \epsilon \otimes \mathrm{id}_Y} Z \otimes Y$$

is equal to a. The composition of the first two arrows equals $(\mathrm{id}_{Z \otimes Y} \otimes \eta) \circ a : Z' \to Z \otimes Y \to Z \otimes Y \otimes X \otimes Y$. Then the composition of $\mathrm{id}_{Z \otimes Y} \otimes \eta$ and $\mathrm{id}_Z \otimes \epsilon \otimes \mathrm{id}_Y$ equals the identity by definition of the dual. Similarly for the other composition. We omit the proof of the second equality. □

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